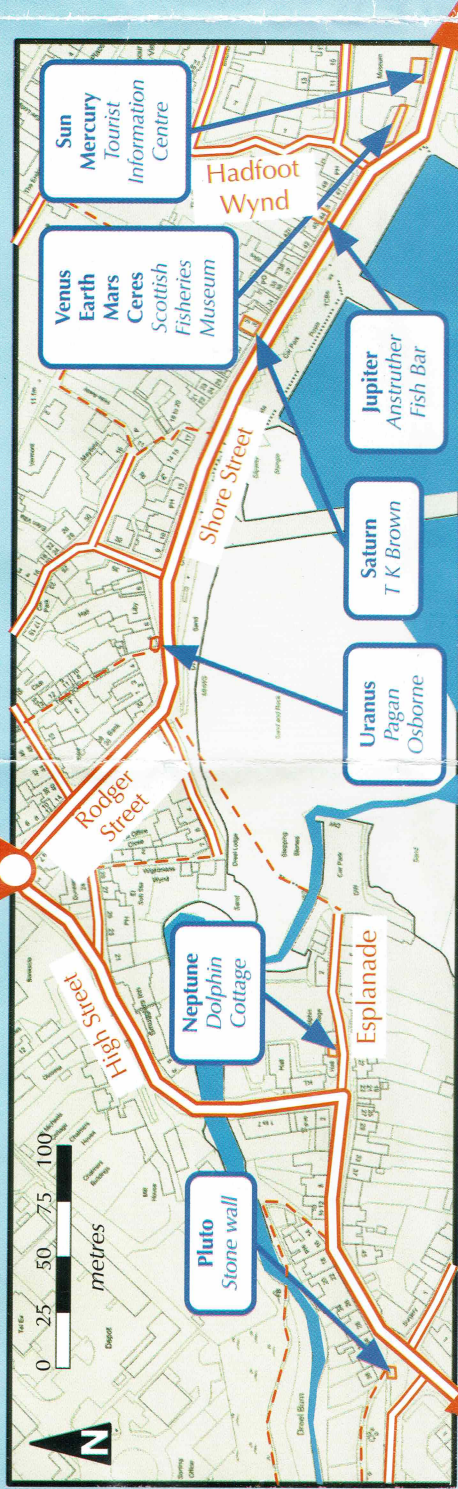


to St Andrews

to Crail



to Pittenweem

to Crail

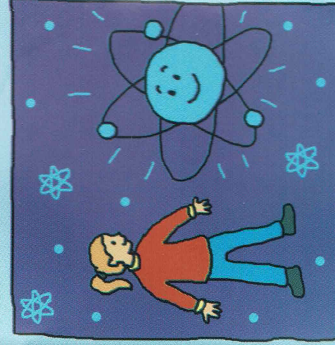
The Anstruther Model Solar System is a true scale model. The Sun and planets, and the distances between them, are all to the same scale. This is why the planets look so small compared to the empty space between them. The scale of the model is 1 to ten thousand million. To get from the real world to the model, divide the real world size by ten thousand million. To get from the model to the real world, multiply the model size by ten thousand million. The smallest thing in the model is 0.1 mm, which is 1 million meters in the real world.

The model is made of 11 bronze plates, mostly fixed to the walls along Shore Street in Anstruther. The Sun plate is at the Tourist Information Office. (It used to be the Sun Tavern — you can still see the name on the wall.) The real planets are not usually in a straight line like this. They are all going around the Sun at different speeds. From time to time a few do roughly line up. It's unusual for this to happen, but the only noticeable effect is that more people start taking an interest in astronomy!

Body	Real world	
	Diameter x 1 000 km	Distance x 1 000 000 km
Sun	1390.0	0.0
Mercury	4.9	57.9
Venus	12.1	108.2
Earth	12.8	149.6
Mars	6.8	227.9
Ceres	0.9	413.9
Jupiter	143.0	778.3
Saturn	120.5	1429.4
Uranus	51.1	2871.0
Neptune	49.5	4504.0
Pluto	2.3	5913.5

* Ceres is too small to see — but there is a bronze plate for it

Scale model	
Diameter mm	Distance m
139.0	0.0
0.5	5.8
1.2	10.8
1.3	15.0
0.7	22.8
*	41.4
14.3	77.8
12.1	142.9
5.1	287.1
5.0	450.4
0.2	591.4



out which element's atom you are closest to in size?

The model includes the Sun and the eight planets from Mercury to Neptune. The dwarf planets Ceres and Pluto are included, though Ceres is too small to see. Each of these has its own plate. Moons are included if they are big enough to see and close enough to fit on the plate. There are some moons which are big enough to see, but too far away from their planet.



What's missing? The whole of the universe outside the Solar System isn't in the model! The nearest star to the Sun is Alpha Centauri. At the scale of this model,



Alpha Centauri is about 4 150 000 m away — about as far away as Cairo, in Egypt.

When the Voyager 1 probe was leaving the Solar System in 1990, scientists made it look back and photograph the planets. One of them was a pale blue dot. Here's what Carl Sagan wrote about the image:

Look again at that dot. That's here. That's home. That's us. On it everyone you love, everyone you ever heard of, every human being who ever was, lived out their lives.



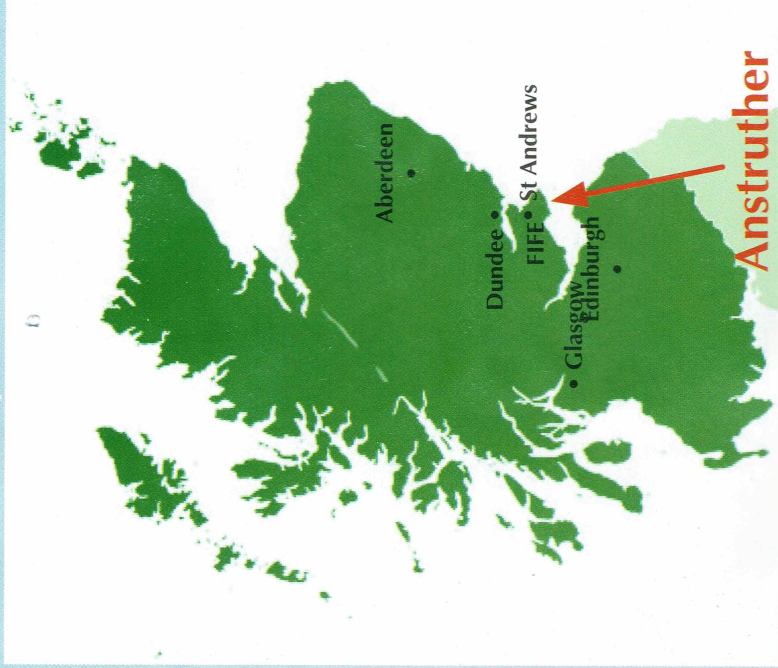
Image: NASA / JPL

The aggregate of our joy and suffering, thousands of confident religions, ideologies, and economic doctrines, every hunter and forager, every hero and coward, every creator and destroyer of civilization, every king and peasant, every young couple in love, every mother and father, hopeful child, inventor and explorer, every teacher of morals, every corrupt politician, every "superstar," every "supreme leader," every saint and sinner in the history of our species lived there — on a mote of dust suspended in a sunbeam. [...] There is perhaps no better demonstration of the folly of human conceits than this distant image of our tiny world. To me, it underscores our responsibility to deal more kindly with one another, and to preserve and cherish the pale blue dot, the only home we've ever known.

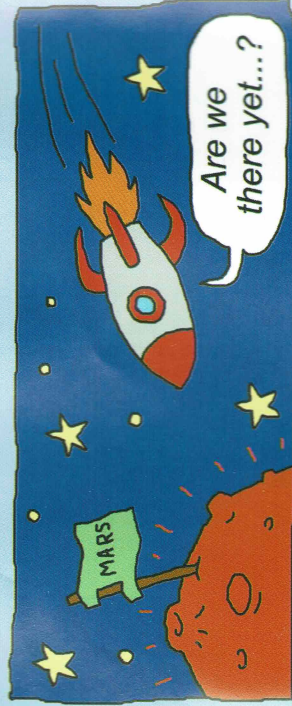
— Pale Blue Dot. Carl Sagan, 1994.



This project was part-financed by the Scottish Government and the European Community Leader in Fife Programme 2007-2013.



In the next 10 or 20 years we may send a human expedition to Mars. It would take about 18 months each way for the journey, even when Mars and Earth are at their closest. The travellers would have to stay a year on Mars until they were close enough to start the return journey. They would have to take enough food, water



and air to last them, unless they can somehow recycle them or make more. They will face dangers that no one has ever known and no one can imagine. Who will they be? The travellers will almost certainly be children who are alive today.

Anstruther Model Solar System

A true-scale model of our Solar System



Reproduced courtesy of Fonestell LLC

— Saturn as seen from Titan —

Be astounded!

by the vastness of the universe

Walk to Pluto and back!
in time for tea

Travel through space!
faster than light

Shrink!

to the size of an atom

Learn!

about our home in space